

Wavefunction and Schrödinger's equation

Wave-particle duality

- All particles can sometimes behave as waves (diffract, interfere) $\rightarrow$ wavepackets.
- Equating the total energy $E=mc^2$ w/ the Planck energy of a quantum $E=hf$, we get

$$E=mc^2=hf$$

For a wave travelling at speed c , $c=f\lambda$, so

$$mc \cdot (f\lambda) = hf$$

$$\lambda = \frac{h}{p}$$

Where $p=mc$ is the momentum of the particle.

- The momentum p can be written as

$$p = \frac{h}{\lambda} = \frac{h}{2\pi\lambda} \cdot \frac{2\pi}{\lambda} = \hbar k$$

The energy E can be written as

$$E = hf = \frac{h}{2\pi} \cdot 2\pi f = \hbar \omega$$

Where $\hbar = \frac{h}{2\pi}$ is the reduced Planck's constant ($h = 6.63 \times 10^{-34}$ Js)

- Wave effects of particles are significant when the distance of obstacles is comparable to its wavelength

The wavefunction ψ .

- The wavefunction ψ contains information on all the measurable parameters of the particle.
- In the 1D case, the wavefunction is $\psi = \psi(x, t)$.
- The probability density function of finding the particle is

$$f(x) = \psi \cdot \psi^*$$

For real $\psi(x, t)$, this becomes

$$f(x) = |\psi(x, t)|^2$$

- The probability of finding a particle b/w $x=a$ and $x=b$ is

$$P(a \leq x \leq b) = \int_a^b |\psi|^2 dx \dots$$

- We can operate on the wavefunction ψ to obtain the parameters of the wave:

$\rightarrow$ Momentum operator $\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}$. so $\hat{p}_x(\psi) = \frac{\hbar}{i} \frac{\partial \psi}{\partial x} = p_x \cdot \psi$.

$\rightarrow$ Energy operator $\hat{E}_x = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$ so $\hat{E}_x(\psi) = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E_x \psi$

- eg: Consider a plane wave $\psi = Ae^{i(kx - \omega t)}$. Find its momentum and energy.

Applying momentum operator $\hat{p}_x$ on the wavefunction ψ ,

$$\hat{p}_x(\psi) = \frac{\hbar}{i} \frac{\partial \psi}{\partial x} = \frac{\hbar}{i} i k A e^{i(kx - \omega t)} = \hbar k \psi = \frac{\hbar}{2\pi} \cdot \frac{2\pi}{\lambda} \psi = p_x \psi.$$

Applying energy operator $\hat{E}_x$ on the wavefunction ψ ,

$$\hat{E}_x(\psi) = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = -\frac{\hbar^2}{2m} (-k^2) A e^{i(kx - \omega t)} = \hbar \omega \psi = \frac{\hbar}{2\pi} \cdot 2\pi f \psi = E_x \psi.$$

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The Schrödinger's equation.

- The kinetic energy of a particle is given by $T = \frac{p^2}{2m}$, so

$$T\psi = \frac{p^2\psi}{2m} = \frac{1}{2m} \cdot \frac{\hbar^2}{i} \frac{\partial}{\partial x} \left(\frac{\hbar}{i} \frac{\partial \psi}{\partial x} \right) = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$$

(The kinetic energy operator $\hat{T} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$ so $\hat{T}\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = T\psi$)

- The potential energy of a particle is given by V ,

- The Schrödinger's eqn. is $\Rightarrow$ simple statement of $T+V=E$, i.e.

kinetic energy + potential energy = total energy

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$$

If the total energy is constant, i.e. $-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi = \text{const}$, then we have the time independent Schrödinger's eqn. (TISE)

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = E\psi$$

- To solve for the arbitrary const in the general sol'n for Schrödinger's eqn, we use the following properties of the wavefunction ψ :

$\rightarrow \psi$ must be continuous (since $|\psi|^2$ is a probability density, which must be continuous)

$\rightarrow \frac{d\psi}{dx}$ must be continuous (o/w, we have infinite $\frac{d^2\psi}{dx^2} \rightarrow$ infinite energy, as $T\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$)

$\rightarrow \int_{-\infty}^{\infty} |\psi|^2 dx = 1$ (probability of finding the particle somewhere is 1)

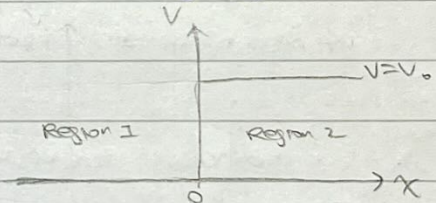
Particle and a potential barrier (infinite width)

- consider the potential energy profile $V(x) = \begin{cases} 0 & x \leq 0 \text{ (region I)} \\ V_0 & x > 0 \text{ (region II)} \end{cases}$

The TISE is:

Region I $-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi$

Region II $-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V_0\psi = E\psi$



- case 1: $E > V_0$.

Region I: $\alpha_1 = \sqrt{\frac{2m}{\hbar^2} E}$ is real $\rightarrow \psi = A_1 e^{i\alpha_1 x} + B_1 e^{-i\alpha_1 x}$

Region II: $\alpha_2 = \sqrt{\frac{2m}{\hbar^2} (E - V_0)}$ is real $\rightarrow \psi = A_2 e^{i\alpha_2 x} + B_2 e^{-i\alpha_2 x}$

leftward transmitted wave is zero as there is no physical reason the wave reflects at $x > 0$.

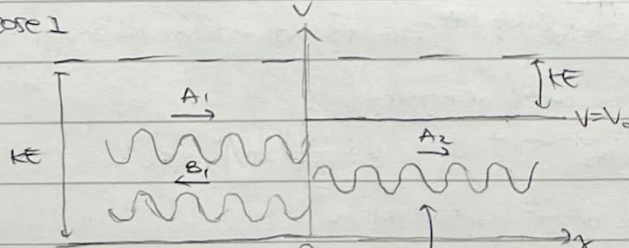
- case 2: $E < V_0$.

Region I: $\alpha_1 = \sqrt{\frac{2m}{\hbar^2} E}$ is real $\rightarrow \psi = A_1 e^{i\alpha_1 x} + B_1 e^{-i\alpha_1 x}$

Region II: $\alpha_2 = \sqrt{\frac{2m}{\hbar^2} (E - V_0)}$ is imaginary $\rightarrow \psi = A_2 e^{\beta_2 x} + B_2 e^{-\beta_2 x}$ where $\beta_2 = i\alpha_2$ is real.

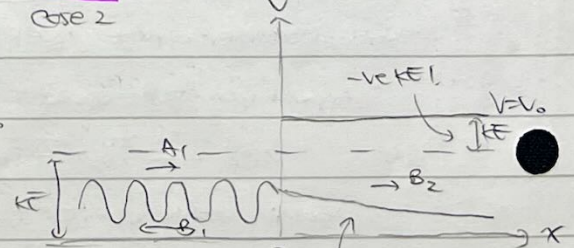
exponential growth term is zero, o/w we have infinite $\psi \rightarrow$ infinite $|\psi|^2$

case 1



transmitted wave has a higher wavelength and less energy.

case 2



tunnelling.

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Particle and a potential barrier (finite width)

- consider the potential energy profile $V(x) = \begin{cases} 0 & x \leq 0 \\ V_0 & 0 < x \leq b \\ 0 & x > b \end{cases}$ (region I), (region II), (region III), and the case $E < V_0$.

The TISE is:

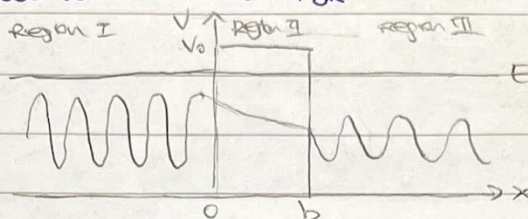
Region I $-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \rightarrow \alpha_1 = \sqrt{\frac{2m}{\hbar^2} E}$ is real $\rightarrow \psi = A_1 e^{i\alpha_1 x} + B_1 e^{-i\alpha_1 x}$

Region II $-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V_0 \psi = E\psi \rightarrow \alpha_2 = \sqrt{\frac{2m}{\hbar^2} (E - V_0)}$ is imaginary $\rightarrow \psi = A_2 e^{\beta_2 x} + B_2 e^{-\beta_2 x}$ [$\beta_2 = i\alpha_2$]

Region III $-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \rightarrow \alpha_3 = \sqrt{\frac{2m}{\hbar^2} E}$ is real $\rightarrow \psi = A_3 e^{i\alpha_3 x} + B_3 e^{-i\alpha_3 x}$ terms are zero due to same reasons as in previous section

- In general, region III intercepts the decaying wave before it reaches 0, and allows it to continue as a travelling wave in region III. $\rightarrow$ wave can tunnel through finite barriers.

- To solve for ψ , we need to ensure that $\psi, \frac{d\psi}{dx}$ is continuous at $x=0, x=b$.



Particle in a box

- consider the potential energy profile $V(x) = \begin{cases} \infty & x \leq 0 \\ 0 & 0 < x \leq L \\ \infty & x > L \end{cases}$ (region I), (region II), (region III)

The TISE is

Region I N/A (particle-wave cannot exist in infinite barrier)

Region II $-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \rightarrow \alpha = \sqrt{\frac{2m}{\hbar^2} E}$ is real $\rightarrow \psi = A e^{i\alpha x} + B e^{-i\alpha x} = A \sin \alpha x + B \cos \alpha x$

Region III N/A (particle-wave cannot exist in infinite barrier)

- ψ must be continuous, so BC are $\psi(x=0)=0$ and $\psi(x=L)=0$.

$\psi(x=0)=0 \rightarrow B=0$

$\psi(x=L)=0 \rightarrow \alpha L = n\pi \rightarrow \alpha = \frac{n\pi}{L} \dots n \in \mathbb{Z}^+$

- $\int_{-\infty}^{\infty} |\psi|^2 dx = 1$, so

$\int_0^L [A \sin(\frac{n\pi}{L} x)]^2 dx = 1 \rightarrow A = \sqrt{\frac{2}{L}}$

$\therefore \psi = \sqrt{\frac{2}{L}} \sin(\frac{n\pi}{L} x)$

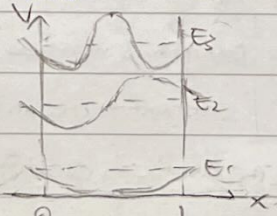
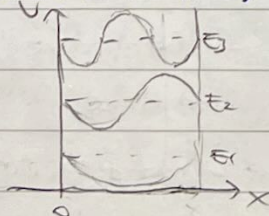
- Applying the kinetic energy operator, we find that

$T\psi = \hat{T}(\psi) = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = -\frac{\hbar^2}{2m} \left(-\frac{n^2\pi^2}{L^2}\right) \psi$

$\therefore T_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \rightarrow$ energy levels are quantised

(In general, for a wave function w wave no. k its kinetic energy is given by $T = \frac{\hbar^2 k^2}{2m} = \frac{p^2}{2m}$)

- If the height of the well is finite, the wavefunction ψ leaks outside the box



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Heisenberg's uncertainty principle.

- Heisenberg's uncertainty principle states that we cannot measure both position and momentum absolutely accurately. If Δx is the error in determining position, Δp is the error in determining momentum.

$$\Delta p \Delta x \geq \frac{\hbar}{2}$$

Equivalently, we cannot measure both energy and time absolutely accurately. If ΔE is the error in determining energy, Δt the error in determining time.

$$\Delta x = v \Delta t \quad ; \quad \Delta E = \frac{dE}{dp} \Delta p = \frac{d}{dp} \left(\frac{p^2}{2m} \right) \Delta p = \frac{2p}{2m} \Delta p = v \Delta p.$$

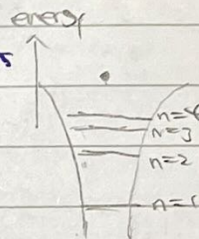
$$\therefore \left(\frac{\Delta E}{v} \right) (v \Delta t) \geq \frac{\hbar}{2}$$

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

Atomic structure

Atoms

- In an atom, the potential energy term accounts for e^-p^+ and e^-e^- interactions
- Consider the case of a H atom. The e^- is bound by a Coulombic potential well (particle in a box, but the box is not rectangular)



$$F = \frac{mv^2}{r} = \frac{q^2}{4\pi\epsilon_0 r^2} \rightarrow mv^2 = \frac{q^2}{4\pi\epsilon_0 r}$$

$$\therefore E = KE + PE = \frac{1}{2}mv^2 - \frac{q^2}{4\pi\epsilon_0 r} = -\frac{q^2}{8\pi\epsilon_0 r}$$

The waves must interfere constructively around the orbit

$$2\pi r = n\lambda$$

$$\text{angular momentum } 2\pi r = n \frac{h}{p}$$

$$mvr = n\hbar$$

Previously, we find that $mv^2 = \frac{q^2}{4\pi\epsilon_0 r}$, so $v = \left(\frac{q^2}{4\pi\epsilon_0 m r} \right)^{1/2}$

$$\therefore m r \left(\frac{q^2}{4\pi\epsilon_0 m r} \right)^{1/2} = \left(\frac{m q^2}{4\pi\epsilon_0} \right)^{1/2} r^{1/2} = n\hbar$$

$$r = \frac{n^2 \hbar^2}{m q^2 / 4\pi\epsilon_0}$$

$$\rightarrow E = -\frac{q^2}{8\pi\epsilon_0 r} = -\frac{m q^4}{22\pi^2 \epsilon_0^2 (n\hbar)^2} \propto \frac{1}{n^2}$$

Chemical bonding.

- C, Si, Ge have 4 valence e^- . All form diamond structure solids. Each atom forms 4 bonds w/ its 4 valence e^- .
- We can also form compounds w/ an average of 4 e^- per atom, for example, in GaAs, Ga has 3 valence e^- , As has 5 valence e^- $\rightarrow$ same structure as diamond, w/ alternate atoms of Ga and As $\rightarrow$ Zincblende lattice
- + The diamond cubic structure is a FCC lattice w/ half of its interstitial holes filled

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crystal planes

- Crystal planes are defined using Miller indices.

- If the Miller indices are (h, k, l) , the plane will cut the x, y, z axis at an integer multiple of $(\frac{1}{h}, \frac{1}{k}, \frac{1}{l})$ (h, k, l are integers).

- e.g.: A plane cuts the x, y, z axis at $2, 4, 1$ respectively.

Taking reciprocals and multiplying by the LCM of the denominators, we get $\frac{1}{2}, \frac{1}{4}, 1 \xrightarrow{\times 4} 2, 1, 4$.

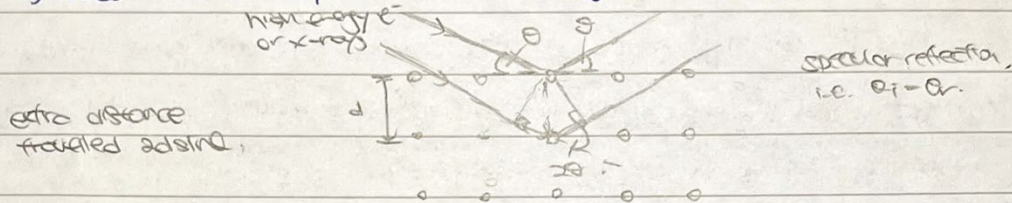
$\therefore$ The plane has Miller indices $(2, 1, 4)$

Bragg's Law - electron diffraction

- The diffraction of e^- can be used to identify properties of a crystal:

$\rightarrow$ 1) impinge a beam of high energy e^- (X-rays) on a crystal at an angle θ .

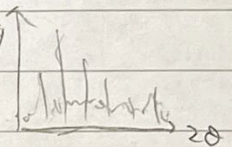
$\rightarrow$ 2) measure the intensity pattern of e^- strikes on a screen.



There is constructive interference when the extra distance travelled is $n\lambda$, i.e.

$$n\lambda = 2d \sin \theta \quad [\text{Bragg's Law}]$$

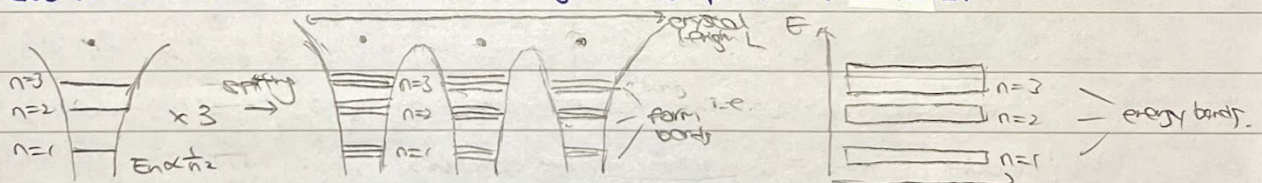
The diffraction pattern has peaks at certain angles $\rightarrow$ identify d



conduction in metals and semiconductors

Formation of energy bands

- As shown previously, atoms have discrete energy levels. If we have n identical atoms together, due to Pauli's exclusion principle, each energy level is split into n sub-levels.

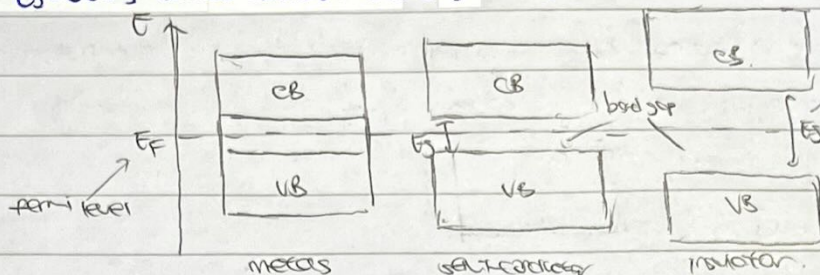


- As n becomes very large, the sub-levels become very close to each other $\rightarrow$ formation of energy bands.

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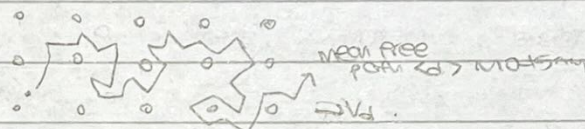
Metals, Insulators and Semiconductors.

- The Fermi level is the thermodynamic work req. to add one e^- to the solid.
- The valence band (VB) and conduction band (CB) are the bands closest to the Fermi level. It is the energy gap E_g b/w VB and CB that determines the electrical conductivity of the solid.
- Metals: $E_g = 0 \text{ eV}$; semiconductors: $0 \text{ eV} < E_g < 2 \text{ eV}$; insulators: $E_g > 2 \text{ eV}$



Conduction in metal

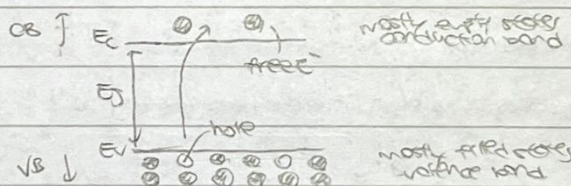
- In metals outer e^- s are shared b/w bands and run around b/w all atoms as free e^- s. The free e^- s are available for conduction while core e^- stay fixed to their atoms.
- The free e^- are like a gas confined to the volume of material, and move randomly around, colliding w/ vibrating host atoms w/ their next free charge.



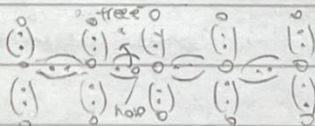
- The conductivity of the metal is determined by the e^- conc. and drift velocity of e^-

Conduction in semiconductors

- Consider the band diagram of a semiconductor



When $T \neq 0 \text{ K}$, some e^- have sufficient thermal energy ($> E_g$) to get promoted from the VB to the CB, becoming a free e^- and leaving a hole in the process.



- The conductivity of the semiconductor is determined by the e^-/h^+ conc (depends on thermal energy and doping) and drift velocity of e^-/h^+

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Carrier concentration in a semiconductor

- For an intrinsic semiconductor (pure, undoped) at thermal eqm, the no. of free carriers per unit volume available for conduction is the intrinsic carrier conc. n_i
- Thermal eqm. implies the material is kept in the dark and no electrical bias is applied, i.e. the only source of energy for e^- is thermal due to ambient temp.
- The intrinsic carrier conc. n_i varies w/ temp. as follows:

$$n_i \propto T^{3/2} e^{-\frac{E_g}{2kT}}$$

obeys Arrhenius law $\exp(-\frac{Q}{2kT})$
→ an activation process

In Si at 300K, $n_i \sim 1.05 \times 10^{10} \text{ cm}^{-3}$.

- At thermal eqm, the no. of e^-/h^+ in the conduction/valence band per unit volume n/p is equal to the intrinsic carrier conc. n_i , i.e.

$$n_i = n = p$$

- e^-/h^+ conc. can be shifted from intrinsic conc. $n/p = n_i$ by doping the semiconductor to get an extrinsic semiconductor. The mass action law states that for any semiconductor,

$$np = n_i^2$$

- N-type doping is adding impurity atoms w/ 5 valence e^- (eg. P) into a Si lattice. 4 e^- participate in bonding and 1 escapes the dopant, and is free to conduct.
- The dopants are donors as they donate e^- . When dopant conc. N_D is much greater than the intrinsic carrier conc. n_i ($N_D \gg n_i$), then

$$n = n_i + N_D \approx N_D$$

→

$$p = \frac{n_i^2}{n} = \frac{n_i^2}{N_D}$$

- P-type doping is adding impurity atoms w/ 3 valence e^- (eg. B) into a Si lattice. 4 e^- are needed to participate in bonding so 1 is pulled in from the lattice → a hole escapes the dopant and is free to conduct.

- The dopants are acceptors as they accept e^- . When dopant conc. N_A is much greater than the intrinsic carrier conc. n_i ($N_A \gg n_i$), then

$$p = n_i + N_A \approx N_A$$

→

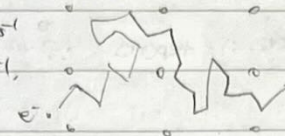
$$n = \frac{n_i^2}{p} = \frac{n_i^2}{N_A}$$

* Typically, at room temp, $n_i \sim 10^{10} \text{ cm}^{-3}$, $N_A/N_D \sim 10^{15} \text{ cm}^{-3}$,

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Velocity of carriers

- When an electric field E is applied, e^-/h^+ start moving to the $\pm$ ve electrode.
- In a semiconductor/metal, the charge carriers cannot accelerate indefinitely, but rather accelerate then immediately scatter (collide w/ lattice / defects / each other). $\rightarrow$ the motion is intermittent bursts of speed and fall to low velocity. (random walk).

$\langle v \rangle \sim 10^6 \text{ m/s}$
 $V_d \sim 10^{-4} \text{ m/s}$


- At low electric fields E , the drift velocity V_d is prop. to the electric field E

$$V_d = \mu E$$

where μ is the mobility.

- ✓ Note that increasing dopant conc. N_A/N_D increases the no. of defects, hence effects of scattering, thus reducing the mean free path and drift velocity V_d .

- At low electric fields E , the drift velocity of e^-/h^+ , $V_{d,n}/V_{d,p}$ is given by

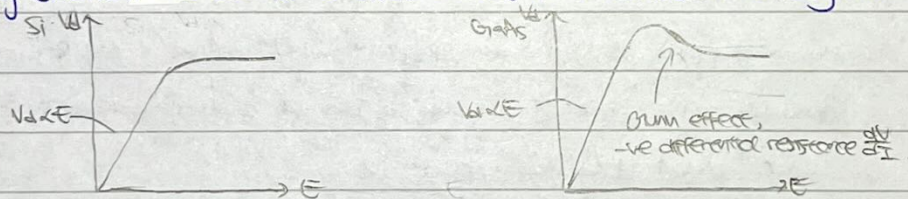
$$V_{d,n} = \mu_n E$$

$$V_{d,p} = \mu_p E$$

Typically, in Si, $\mu_n = 1500 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$; $\mu_p = 500 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.

- At high electric fields E , the drift velocity depends on the semiconductor. In most semiconductors inc. Si, the velocity saturates at high fields.

- However, in GaAs, we observe the Gunn effect — the velocity initially increases, then reduces w/ increasing electric field E due to the effective mass of the carriers m^* increasing + mobility μ decreasing. due to inter-valley transfer in E-K diagram



Mobility and effective mass.

- In a crystal lattice, the charge carrier interacts w/ the lattice — it polarises the neighbouring atoms and moves through the lattice. To use Newton's laws to model the charge carrier, we must use its effective mass m^* instead of its rest mass m .

- consider a charge carrier moving through a crystal lattice due to an electric field.

$$\frac{dp}{dt} = \frac{m^* V_d}{\tau} = qE = F$$

$$V_d = \frac{q\tau}{m^*} E \rightarrow \mu = \frac{q\tau}{m^*}$$

where τ is the mean time between collisions

- The effective mass m^* is related to the curvature of the band on a E-K plot.

$$E = \frac{p^2}{2m^*} = \frac{\hbar^2 k^2}{2m^*} \rightarrow \frac{\partial^2 E}{\partial k^2} = \frac{\hbar^2}{m^*} \rightarrow m^* = \frac{\hbar^2}{\frac{\partial^2 E}{\partial k^2}}$$

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Drift current in a semiconductor

- The drift current density J_{dr} is given by

$$J_{dr} = qnV_{dr} + qpV_{dp} = qn\mu_n E + qp\mu_p E = q(n\mu_n + p\mu_p)E$$

By def'n of current density,

$$I = J_{dr}A = q(n\mu_n + p\mu_p)AE$$

Relating E and V using $E = \frac{dV}{dx}$,

$$I = q(n\mu_n + p\mu_p)A \frac{dV}{dx}$$

$$\int_0^L I dx = \int_0^V q(n\mu_n + p\mu_p)A dV$$

$$V_a = \frac{1}{q(n\mu_n + p\mu_p)} \frac{L}{A} I$$

$$\therefore R = \frac{V_a}{I} = \frac{1}{q(n\mu_n + p\mu_p)} \frac{L}{A} \rightarrow \rho = \frac{A}{L} R = \frac{1}{q(n\mu_n + p\mu_p)}$$

- If the semiconductor is intrinsic, $n_i = n = p$,

$$\rho = \frac{1}{q(n\mu_n + p\mu_p)} = \frac{1}{q n_i (\mu_n + \mu_p)}$$

If the semiconductor is doped w/ donor dopants of conc. N_D , $N_D \gg n_i \rightarrow n \approx N_D$, $p \approx \frac{n_i^2}{N_D}$ ($n \gg p$)

$$\rho = \frac{1}{q(n\mu_n + p\mu_p)} \approx \frac{1}{q(n\mu_n)}$$

If the semiconductor is doped w/ acceptor dopants of conc. N_A , $N_A \gg n_i \rightarrow p \approx N_A$, $n \approx \frac{n_i^2}{N_A}$ ($p \gg n$)

$$\rho = \frac{1}{q(n\mu_n + p\mu_p)} \approx \frac{1}{q(p\mu_p)}$$

Electrostatics in semiconductors

Poisson's equation.

- starting from the differential form of Gauss' law for electrostatics,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon}$$

Denoting the potential as ϕ , and using $\mathbf{E} = -\nabla\phi$, we have

$$\nabla^2 \phi = -\frac{\rho}{\epsilon}$$

In 1D, this simplifies to

$$\frac{d^2\phi}{dx^2} = -\frac{\rho}{\epsilon}$$

Electrostatics in insulators.

- Consider an insulator in a // plate capacitor. The insulator has no charges, one end ($x=0$) is at potential V_a while the other end ($x=L$) is at 0 potential.

Gauss law: $\frac{dE}{dx} = \frac{\rho}{\epsilon} = 0 \rightarrow E = q = \text{const}$

$$-\frac{d\phi}{dx} = E \rightarrow \phi = q x + C_1$$

BC: $\phi(x=0) = V_a, \phi(x=L) = 0 \rightarrow E = \frac{V_a}{L}, \phi = V_a(1 - \frac{x}{L})$

Electrostatics in semiconductors

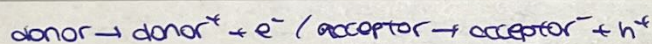
- Generally in semiconductors, there are two types of charges,

↳ 1) mobile charges — e^-, h^+ .

They will respond to any $\mathbf{E}$ field and move about. Therefore their conc. will depend on the potential in the semiconductor

↳ 2) fixed charges — dopant ions donor $^+$ /acceptor $^-$

The n-type donor / p-type acceptor dopant has given up its e^-/h^+ and therefore is a fixed charge.



- The semiconductor is still neutral after doping.

- Sometimes, a region in a semiconductor might be depleted — it is nearly void of carriers. If the depletion region is doped, the only charges present will be the dopant ions.

- Consider a p-doped semiconductor w/ acceptor dopant conc. N_A . The region from $x=0$ to $x=d$ in the semiconductor is depleted.

charge density:

$$\rho = q(N_A - n) = -qN_A$$

Gauss law:

$$\frac{dE}{dx} = \frac{\rho}{\epsilon} = -\frac{qN_A}{\epsilon} \rightarrow E = -\frac{qN_A}{\epsilon}x + C_0$$

$$-\frac{d\phi}{dx} = E \rightarrow \phi = \frac{qN_A}{\epsilon} \frac{x^2}{2} - C_0x + C_1$$

BC: $E(x=d) = 0 \rightarrow E = \frac{qN_A}{\epsilon}(d-x), \phi_{0 \rightarrow d} = -\frac{qN_A}{\epsilon} \frac{d^2}{2}$

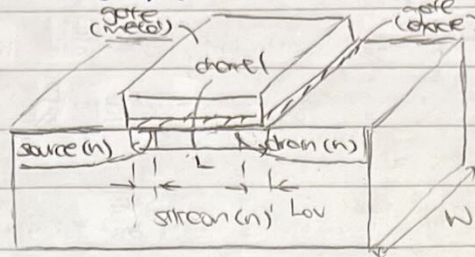
 $\propto \sqrt{\phi}$

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MOSFETS

MOSFET

- A MOSFET has a gate (G), source (S), drain (D) and body contact (connects to the bulk Si)
- The physical construction of a (n-type) MOSFET is as follows

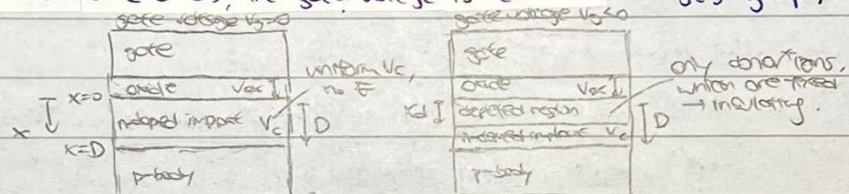


L is the channel length, W is the channel width, L_{ov} is the overlap length, $\frac{W}{L}$ is the aspect ratio

- The gate voltage V_g controls the resistance of the channel via field effect. The transistor is ON/OFF when the channel has low/high resistance.
- For depletion mode MOSFET, a -ve V_T is req. at G_1 to turn the transistor OFF (o/w it is ON)
- For enhancement mode MOSFET, a +ve V_T is req. at G_1 to turn the transistor ON (o/w it is OFF)

Depletion-mode MOSFET calculation.

- consider a n-type depletion-mode MOSFET. Assume the entire n-doped implant is at potential V_c , and a voltage V_{ox} is dropped across the oxide. Let the thickness of the depletion region be x_d . When full depletion occurs ($x_d = D$), the gate voltage is the pinch-off voltage, $V_g = V_p$.



consider the depletion region

charge density.. $\rho = q(N_D + P^+ - N_A^+ - P^-) = qN_D$

Gauss Law:

$$\frac{dE}{dx} = \frac{\rho}{\epsilon} = \frac{qN_D}{\epsilon} \rightarrow E = \frac{qN_D}{\epsilon}x + C_1$$

$$-\frac{d\phi}{dx} = E \rightarrow \phi = -\frac{qN_D}{\epsilon}\frac{x^2}{2} - C_1x + C_2$$

BC: $E(x=x_d) = 0 \rightarrow E = \frac{qN_D}{\epsilon}(x-x_d), \phi_{0 \rightarrow d} = \frac{qN_D}{\epsilon}\frac{x_d^2}{2}$

BC: $\phi(x=x_d) = V_c \rightarrow \phi(x=0) = \phi(x=x_d) - \phi_{0 \rightarrow d} = V_c - \frac{qN_D}{\epsilon}\frac{x_d^2}{2}$

$$\therefore \phi(x=0) = V_g - V_{ox} = V_c - \frac{qN_D}{\epsilon}\frac{x_d^2}{2}$$

Full depletion when $x_d = D, V_g = V_p$

$$\therefore V_p = V_c + V_{ox} - \frac{qN_D}{\epsilon}\frac{D^2}{2}$$

* Actually, the thickness of the depletion region is not uniform over the channel length since

$V_{gd} < V_{gd}$ (calculation beyond the scope of this course).

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Transit time and maximum current

- At low fields, $V_d = \mu E$.

At high fields, velocity saturates and equals the scattering limited velocity $V_d = V_s = 10^5 \text{ ms}^{-1}$ (for Si)

- The transit time t is then given by

$$t = \frac{L}{V_d}$$

and the max. (saturation) current is given by

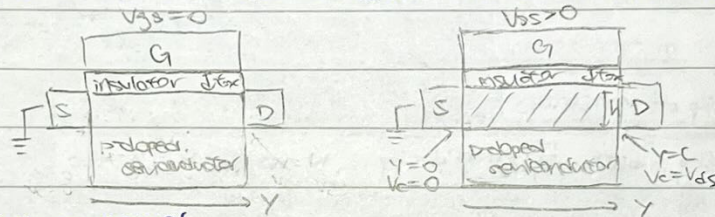
$$I_{\text{sat}} = \frac{Q}{t} = \frac{qN_0(WDL)}{t} = qN_0(WD)V_s$$

Thin film transistors

Thin film transistors (TFTs)

- TFTs are built w/ depositing thin films on substrates (eg glass, plastic), and are the building blocks of large area electronic systems (eg. displays)

- The I-V characteristics of TFTs are similar to enhancement mode MOSFETs, but w/ some diff. due to the nature of the material. (we will ignore these diff. in this course)



- At $V_{gs} = 0$, there is no channel

- At $V_{gs} > V_T$ (threshold voltage), there is a conductive channel of e^- b/w source and drain. When a channel forms, it has a channel potential $V_c(y)$ that varies from source to drain

- If we imagine the channel (sheet of e^-) to have a thickness h , the current through the channel I_{ds} is

$$\frac{I_{ds}}{Wh} = -qnV_{dn} = -qn\mu_n E = qn\mu_n \frac{dV_c}{dy}$$

- If we define the insulator capacitance per unit area as $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$,

$$Q = qn(LWh) = C_{ox}(LW)(V_{gs} - V_T - V_c)$$

$$qn = \frac{C_{ox}(V_{gs} - V_T - V_c)}{h}$$

$$\therefore I_{ds} = qn\mu_n \frac{dV_c}{dy} (Lh) = \mu_n C_{ox} (V_{gs} - V_T - V_c) \frac{dV_c}{dy}$$

$$\int_0^L I_{ds} dy = \int_0^{V_{ds}} \mu_n C_{ox} (V_{gs} - V_T - V_c) dV_c$$

$$I_{ds} = \frac{\mu_n C_{ox}}{L} \left[(V_{gs} - V_T) V_{ds} - \frac{V_{ds}^2}{2} \right] \quad V_{ds} < V_{gs} - V_T$$

The above operation is only true if $V_{ds} < V_{gs} - V_T$ (linear operation) $\rightarrow$ resistor

- If $V_{ds} \geq V_{gs} - V_T$, there will be a region near the drain where the channel cannot exist ("pinch off")

The current saturates and becomes independent of V_{ds} (saturation operation) $\rightarrow$ gate-controlled current source.

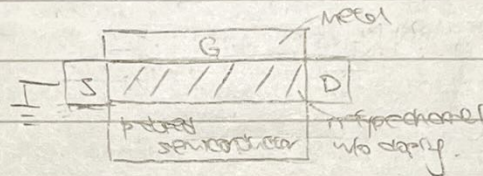
- To calculate I_{ds} in saturation operation, we substitute $V_{ds} = V_{gs} - V_T$

$$I_{ds} = \frac{\mu_n C_{ox}}{2L} (V_{gs} - V_T)^2 \quad V_{ds} \geq V_{gs} - V_T$$

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Metal semiconductor field effect transistor (MESFET).

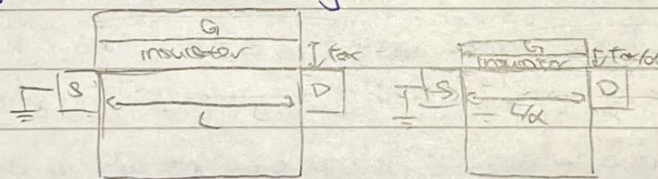
- The key diff. btwn MESFET and MOSFET is that there is no insulator btwn gate metal & semiconductor.
- When a metal (M) interfaces w/ a semiconductor (S), there are two types of contact to be made:
 - ↳ (i) ohmic contact
 - ↳ (ii) schottky / rectifying contact.
- An ohmic contact will ensure the M-S junction behaves like a resistor — there is no charge transfer, this junction is used if we want to inject current from metal to semiconductor, and vice versa.
- A schottky junction forms a diode-like contact — a depletion region forms where both the metal and semiconductor are depleted of free carrier.
- A MESFET uses a gate metal that forms a schottky contact w/ the semiconductor, the resulting depletion region can be controlled to set the source-drain current.
- The MESFET is often used for high freq. operations (e.g. radar) since there is no doping of semiconductor, which would o/w introduce dopant atoms, thus defects, which lowers the mobility μ .
- A disadvantage of the MESFET is that we can't turn it off (like a MOSFET).



Scaling and material of transistors

Scaling of transistors

- scaling implies that we are reducing L, W, t_{ox} by a factor of α .
(practically, scaling down t_{ox} increases tunneling current $\rightarrow \uparrow I_{er}$ and keep t_{ox} const)



- There are two methods for scaling:

↳ (i) constant field scaling.

keep field const $\rightarrow$ scale down applied voltage $\rightarrow$ voltage levels not back-compatible

↳ (ii) constant voltage scaling

keep voltage const $\rightarrow$ field becomes large $\rightarrow$ may cause breakdown

- The following quantities are scaled as follows (L, W, t_{ox} scaled by $\frac{1}{\alpha}$)

Parameter	field	voltage	gate capacitance	oxide capacitance	current	power
symbol	$\mathcal{E}$	V	C_m	C_{ox}	I	P
const-field scaling	1	$\frac{1}{\alpha}$	$\frac{1}{\alpha}$	α	$\frac{1}{\alpha}$	$\frac{1}{\alpha^2}$
const-voltage scaling	α	1	$\frac{1}{\alpha}$	α	α	α

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Materials of transistors

- Si: Best for general purpose MOSFET

It is a v. well established technology $\rightarrow$ low cost

Easy to get high purity wafers w/o defects

- GaAs: Very good for high frequency applications $\rightarrow$ MESFETs

Gunn effect $\rightarrow$ negative differential resistance $\rightarrow$ circuits w/ high Q factors

crystals are quite defective

- GaN: very high breakdown voltage $\rightarrow$ high voltage / high power / high temp applications.

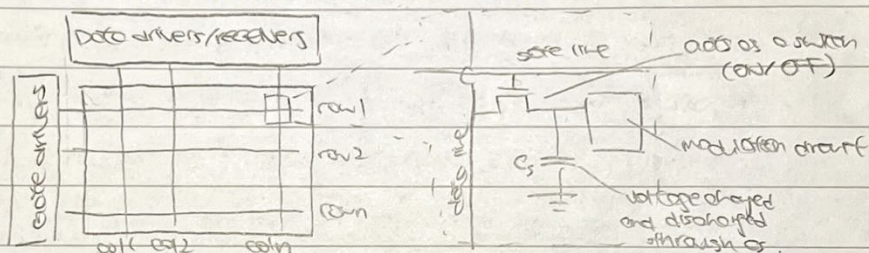
wide band gap semiconductor $\rightarrow$ optoelectronic materials for short wavelength.

Displays

Active matrix displays

- An active matrix display consists of pixel control (transistors + light modulators).

- The architecture for an active matrix display works as follows:



Liquid crystals (LC)

- LC is a display element based on modulating backlight (backlight)

- LCs are a phase of matter between liquid and crystal.

- When the LC is placed along two "aligning guide" crystals, it will direct itself along the guide.

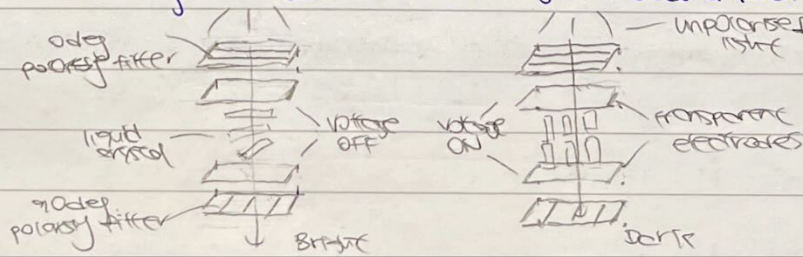
We can use this to obtain a twist between two $\perp$ polarisers.

- When an E-field is applied, it reorients the LCs so no longer twist the light.

- A backlight is modulated as follows:

$\rightarrow$ Voltage OFF: unpolarised backlight $\rightarrow$ polarised to 0 deg $\rightarrow$ slowly twisted by the LC to polarised 90 deg. $\rightarrow$ passes through the second polariser

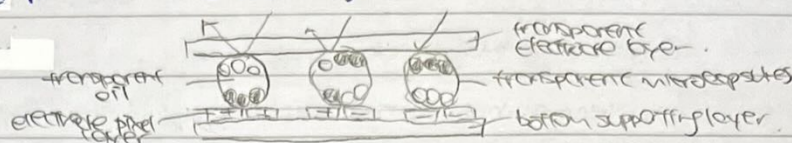
$\rightarrow$ Voltage ON: unpolarised backlight $\rightarrow$ polarised to 0 deg $\rightarrow$ not twisted by the LC and remains at 0 deg $\rightarrow$ does not pass through the second polariser



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E-ink

- E-ink is a display element based on modulating ext. light (reflected ambient light)
- E-ink consists of $\text{TiO}_2/\text{SiO}_2$ charged white/black pigments enclosed in transparent microcapsules, which are sandwiched between transparent conductive electrodes.
- Depending on the field, we have a diff. proportion of white/black particles on the top, which implies pdr/gnd reflection of ambient light.



Light-emitting diodes (LED)

- LED is a display element based on generating light
- LEDs use pn junction diodes. When the diode is forward biased, the built-in potential barrier is lowered and carriers can diffuse across the junction $\rightarrow$ diffusion current.

$$I = I_0 \left(e^{\frac{qV}{kT}} - 1 \right)$$

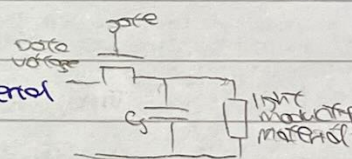
(transport of charge due to a conc. gradient, i.e., diffusion)

- The diffusion of e^-/h^+ from n/p to p/n side leads to an increase in recombination of e^- and h^+ , producing light $\rightarrow$ light intensity depends on the current.
- Direct bandgap semiconductors are preferred.

Pixel circuit

① Voltage programmed pixel circuit (LED, E-ink)

- 1 TFT needed, used as a switch
- The "Data voltage" is the voltage req. to modulate the light-emitting material
- The resistance of the switch is decided by the gate voltage V_g .
- The switch operates in linear bias (resistor) $\rightarrow R_{TFT}$



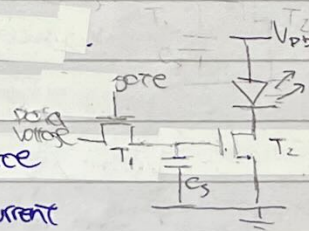
$$R_{TFT} = \frac{dV_g}{dI_{ds}} \approx \frac{1}{\mu C_{ox} (W/L) (V_{gs} - V_t)}$$

in linear mode, $V_{gs} - V_t \rightarrow V_{ds}$

The time const. is therefore $\tau = R_{TFT} C_s$

② Current programmed pixel circuit (LED)

- 2 TFTs needed, T_1 used as a switch, T_2 used as a current source
- The "Data voltage" is the gate voltage req. for T_2 to drive the right current
- T_1 operates in linear bias (resistor); T_2 operates in saturation mode (voltage controlled current source)



$$I_{ds} = \frac{\mu C_{ox} (W/L)}{2} (V_{gs} - V_t)^2$$